



AirPassengers Forecasting Project

About the Project

The AirPassengers Forecasting Project is a Time Series Analysis project that focuses on analyzing and predicting airline passenger traffic using historical data from 1949 to 1960. The project aims to identify trends, seasonal patterns, and growth in passenger demand over time.

Various time-series techniques such as data visualization, seasonal decomposition, moving averages, and ARIMA forecasting are applied to understand passenger behavior and generate future demand predictions. The model's performance is evaluated using RMSE (Root Mean Squared Error) to measure forecasting accuracy.

This project demonstrates the practical application of Time Series Analysis in demand forecasting, business planning, and decision-making. The insights generated from the analysis can help airlines optimize capacity, improve resource allocation, and prepare for future passenger growth.

Objectives

- Analyze historical airline passenger data
- Identify trends and seasonal patterns
- Apply moving average techniques
- Build and train an ARIMA forecasting model
- Predict future passenger demand
- Evaluate model performance using RMSE

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Statsmodels
- Scikit-learn
- Google Colab

Key Features

- Time Series Visualization
- Trend Analysis
- Seasonal Decomposition
- Moving Average Analysis
- ARIMA Forecasting
- Model Evaluation

- Business Insights and Recommendations

Author

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